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A lattice-based forward secure IBE scheme for Internet of things

Renjie Jin [a], Longjiang Qu [a], Rongmao Chen [b,*], Zhichao Yang [c], Yi Wang [b]

[a] School of Sciences, National University of Defense Technology, 410073, Changsha, Hunan, China
[b] School of Computer, National University of Defense Technology, 410073, Changsha, Hunan, China
[c] Department of Information Security, Naval University of Engineering, 430032, Wuhan, Hubei, China

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ABSTRACT

The Internet of Things (IoT) is gaining popularity as it can effectively connect the real world with the Internet. However, security and privacy issues are widely considered the major obstructions to its widespread adoption. Hence, various cryptographic tools (e.g., encryption and signature schemes) have been proposed to build secure and authenticated channels for data communication in IoT. In this work, we investigate the well-known encryption scheme called identity-based encryption (IBE) for IoT-oriented applications. In particular, we are mainly interested in forward-secure IBE (fs-IBE) which could still protect the data transferred in the past when the current secret key stored in the IoT device is stolen. To this end, we propose an fs-IBE scheme which, as far as we know, is the first construction of fs-IBE that can resist quantum attacks. Our scheme is based on the lattice-based hardness assumption namely Learning with Error (LWE), which is widely adopted for designing other quantum-resistant cryptographic schemes. At the core of our design is the minimal cover mechanism in the context of binary tree and we rigorously prove the security of our scheme in the forward-secure, selective ID, chosen-plaintext attack (CPA) model. Besides the post-quantum security, our scheme enjoys comparatively compact ciphertext and secret key, and thus it is suitable for bandwidth-limited IoT communication.

1. Introduction

The Internet of Things (IoT) is widely considered a useful tool to connect the real world and the Internet broadly [4,23], and it provides an effective platform for improving the efficiency of manufacturing and developing powerful applications. IoT-based devices are tiny, sophisticated, networkable electronics embedded into a real-world object which act as a vehicle for connecting the Internet to the real world. The security of data transmission should be taken into consideration when IoT devices are deployed. As it turns out, several security algorithms have been proposed for the goal of secure communication between IoT-based devices. In particular, an efficient secure channel is often established between devices by utilizing cryptographic algorithms, among which public-key tools (e.g., public-key encryption and digital signature) are widely adopted for data encryption and device authentication. While public-key cryptography is a useful tool for secure communication in IoT, it usually involves costly certificate management which causes burden for resource-restricted IoT devices. To eliminate the certificate management, identity-based encryption (IBE) [25,7] could be a desirable encryption tool for IoT systems. In such an application scenario, the public key of an IoT device is its unique identity, and the private key for each device is generated by a private key generator (PKG). Thus, each device does not need to apply for digital certificates, avoiding the tedious process of certificate issuance, revocation, verification and storage [14,27]. Unfortunately, although

* Corresponding author.
E-mail address: chromao@nudt.edu.cn (R. Chen).

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IBE could be adopted to effectively protect the communication in IoT, most of the current constructions of IBE schemes still suffer from the following security issues.

- *Secret Key Exposure*. Cryptosystems provide security guarantees only if the secret key is not compromised. Unfortunately, IoT devices suffer from certain vulnerabilities that could provide entry points for attackers to corrupt the IoT devices and steal the secret key. In particular, it is known that side channel attacks could be launched on IoT devices to recover the secret key by monitoring the device's power consumption, radio emanations, and even the sounds coming from the embedded electronic components [15,31].
- *Quantum Attacks*. Besides the secret key exposure, another threat against IBE schemes is the quantum attack. It is known that IBE built on classical hardness assumptions will be completely broken by quantum computer. Precisely, it has been shown that some typical mathematical problems (e.g., factoring large integers) could be solved in polynomial time via running specific quantum algorithms [26].

In this work, we intend to explore effective approaches that can protect IBE schemes against the aforementioned attacks. Below we first briefly discuss some possible solutions. As a potential countermeasure to the secret key exposure, one could let each IoT device update the identity as well as the secret key periodically. This, however, brings huge computational overhead for the PKG, as the PKG needs to calculate the secret key for the new identity and establish a channel for the key distribution. In order to resolve this issue, *identity-based encryption with forward security* (fs-IBE) is a desirable choice as it supports secret key update without interacting with the PKG. Specifically, for an fs-IBE, the PKG only needs to send the key used for decryption to each IoT device at the initial moment, and each device can update the key locally and periodically without changing its identity (and thus no additional interaction with PKG is required). Hence, the compromise of the current secret key does not have an impact on the security of previously encrypted messages. To defend quantum attacks, one could turn to post-quantum cryptographic techniques for designing quantum-resistant IBE schemes.

Specifically, there exist several hardness assumptions believed to hold in the era of quantum computing. These at least include lattice-based and code-based hardness assumptions. Currently, it is widely accepted that cryptographic schemes designed based on hard lattice problems exhibit high efficiency and low storage costs [18]. Regrettably, to our knowledge, existing forward-secure IBE (fs-IBE) schemes [29,21,22] are constructed based on traditional hardness assumptions.

Given these observations, the primary focus of our work is to design IBE schemes with forward security using lattice-based assumptions. To provide further clarity on our contributions, we delve into the technical challenges associated with constructing fs-IBE schemes based on lattice assumptions.

Technical Challenges. Designing fs-IBE requires that the secret key update should be one-wayness, which is difficult to achieve through lattice as the structure of lattice is not friendly to support update mechanism. Therefore, it is necessary to introduce certain structures that can connect the lattice and fs-IBE to implement the update mechanism of the secret key. Moreover, in the forward-secure CPA model, we allow the adversary query for the secret key $sk_{id^*,j}$, where id^* indicates the challenge identity and j is a random interval after the challenge interval. Thus, it should be guaranteed that such an ability will not provide any additional information about the key generated in previous interval to the adversary, otherwise the adversary might win the attack game trivially.

It is worth noting that Shweta et al. [1] proposed an update algorithm called *BasisDel* for trapdoors of lattice and applied it to the construction of lattice-based HIBE schemes. Intuitively, the underlying idea of updating lattice trapdoors may shed some light on achieving our goal of fs-IBE. Unfortunately, we found that the idea can not be used in our construction since the technique has structural defect in the fs-IBE model. More precisely, they proposed a one-way lattice basis delegation technique. However, each derived lattice basis experiences a certain norm increase compared to the previous lattice basis, indicating a decrease in the quality of the lattice basis. While their technique is suitable for the structure of HIBE, it may not necessarily be suitable for the structure of fs-IBE. A possible reason for this could be that HIBE does not require the secret key to be derived as many times as fs-IBE, which requires more epochs.

Our Results. By introducing the binary tree and the *minimal cover mechanism* [19] into the lattice-based IBE construction, we proposed the first lattice-based fs-IBE. We realized that the application of the minimal cover mechanism in the binary tree would bring an efficient one-way delegation mechanism. To some extent, this is similar to the lattice-based delagation mechanism of Shweta et al. In the concrete scheme, by exploiting the minimal cover mechanism of a binary tree, forward security is achieved. The secret key is associated with a pair (id, t) (id is an identity, t refers to an epoch), and required updated at each interval. In our scheme, we divide the secret key into two parts of which the first part controls the updates while the latter part is used to decrypt the ciphertext. This paper is briefly summarized below in terms of its contributions.

- We present the first lattice-based fs-IBE scheme, and therefore is believed to be secure against quantum attacks. Moreover, under our proposed forward-secure, selective ID, chosen plaintext attack (fs-sID-CPA) model, we give a formal security reduction. Furthermore, we show that the scheme can achieve fs-sID-CCA (resp. IND-CCA) security through FO transformation [16] (resp. "BCHK" transformation [8]).
- We introduce the binary tree into the construction of fs-IBE, and the update of secret key is achieved through the minimal cover mechanism associated with the binary tree. Moreover, we use the lattice basis extension algorithm [13] to connect lattice with fs-IBE. Our structural approach could potentially provide a novel perspective for the design of cryptographic primitives that necessitate secret key derivation.

- Our scheme is characterized by comparatively compact parameter sizes. Specifically, the sizes of the public parameters and the master secret key are independent of the total epochs N, that is, they are of "constant size". The sizes of the secret key and ciphertext exhibit a logarithmic relationship with the period N. Moreover, the operations of encryption/decryption of our scheme are simple computations on lattice, which guarantees a high efficiency when the scheme is used in post-quantum secure IoT.

Other Related Work. There are some fs-PKE schemes that were previously known. The first non-interactive PKE scheme with forward security was presented by Canetti et al. [11]. Moreover, based on the work of Gentry and Silverberg [17], this scheme achieves forward-secure CPA security in standard model. In [21], the authors proposed an fs-PKE scheme which is based on the idea of *hierarchical* IBE (HIBE) [10]. Compared with [11], this scheme has reduced the size of main parameters and hence enjoy a higher efficiency. In [28], the authors defined the model of fs-HIBE for the first time and presented the first fs-HIBE, as well as showing some applications of fs-HIBE.

2. Preliminaries

In this section, we first introduce the syntax and the security model of fs-IBE. Then we briefly introduce the basic definitions and relevant results of lattice, Discrete Gaussian and Learning with Errors.

2.1. Notations

- Denote by $\mathbb{Z}$ the integer ring and $\mathbb{Z}_q$ the ring $\mathbb{Z}/q\mathbb{Z}$, where q is a positive integer.
- Let $\mathcal{C}$ be a probabilistic poly-time (PPT) algorithm. By $\mathbf{y} \leftarrow \mathcal{C}(\mathbf{x})$ we mean the random variable $\mathbf{y}$ is defined as the output of $\mathcal{C}$ with input $\mathbf{x}$ and a coin chosen at random.
- Write $\mathbf{s} \hookleftarrow \mathcal{D}$ if $\mathbf{s}$ is sampled uniformly from distribution $\mathcal{D}$.
- Let $\mathbf{y} \leftarrow_{\$} \mathbb{Z}_q^n$ denote the random vector uniformly chosen from $\mathbb{Z}_q^n$.
- $||\cdot||$: For a vectors set $S := \{\mathbf{s}_0, \cdots, \mathbf{s}_{k-1}\}$ in $\mathbb{R}^m$, we define $||\mathbf{s}_i||$ as the ℓ_2 norm of $\mathbf{s}_i$ $(0 \le 1 \le k-1)$ and write $||S|| := \max_{i\in\{0,\cdots,k-1\}} ||\mathbf{s}_i||$, which is the norm of S.
- Suppose $\widetilde{S} := \{\widetilde{\mathbf{s}}_1, \cdots, \widetilde{\mathbf{s}}_k\}$ is the Gram-Schmidt orthogonalization of $\mathbf{s}_1, \cdots, \mathbf{s}_k$ with respect to the same order. Denote by $||\widetilde{S}||$ the Gram-Schmidt norm of $\widetilde{S}$.

2.2. Fs-IBE: syntax

The definition of fs-IBE is adapted from [28]. An fs-IBE scheme consists of the following five algorithms together with a message space $\mathcal{M}$ and a ciphertext space $\mathcal{C}$:

Setup: Input parameters for the **PKG** are security parameter λ and N (the total periods of forward-secure cryptosystem). The algorithm outputs a pair (*params*, *msk*) where *params* are public parameters, *msk* is the master key and kept secret by **PKG**.

KeyGen: This algorithm takes predefined *params*, *msk* and an identity id, produces the initial secret key of id which corresponds to the first interval of forward-secure system, denoted as $sk_{id,0}$.

Update: This algorithm takes *params* and $sk_{id,i}$ which corresponds to the pair (id, i) (an id and an interval i), returns a secret key $sk_{id,i+1}$ that will be used in interval $i+1$, and subsequently deletes $sk_{id,i}$.

Encrypt: This algorithm takes *params*, a pair (id, i), and a message $M \in \mathcal{M}$, returns a ciphertext $C \in \mathcal{C}$.

Decrypt: This algorithm takes *params*, a ciphertext $C \in \mathcal{C}$ and $sk_{id,i}$, returns either a decrypted message $M \in \mathcal{M}$ or a symbol $\perp$ which represents rejection.

Encryption and decryption must have consistency and strict correspondence. That is, if $sk_{id,i}$ is generated by the algorithm **KeyGen** or **Update** for the pair (id, i), then for all $M \in \mathcal{M}$, the ciphertext C is decrypted via *params* and $sk_{id,i}$ generates the original message M, where C is exactly the ciphertext of encrypting M under the pair (id, i).

2.3. Fs-IBE: security

Security model is adapted from [28].

Forward-secure selective-ID CCA (fs-sID-CCA): We claim that an fs-IBE system satisfies the selective ID, chosen ciphertext attack security if there does not exist a PPT adversary who could win the following attack game with non-negligible advantages.

Init: The adversary submits an id^*. The challenger runs **Setup** and sends the obtained *params* to adversary.

Phase 1: The adversary will make q_i queries for $i = 1, \cdots, m$, and the challenger responds as follows:

- Extraction query (id, j): Run **KeyGen** and **Update** algorithms to generate a secret key $sk_{id,j}$ which corresponds to the pair (id, j), and relays it to adversary.
- Decryption query (id, C, j): Run **KeyGen** and **Update** algorithms to generate a secret key $sk_{id,j}$, and using **Decrypt** to decrypt the submitted ciphertext C, then sends either a message M or a symbol $\perp$ which means rejection.

Challenge: After the adversary finishes the first stage of query, he submits two messages $M_0, M_1 \in \mathcal{M}$ with the same lengths, along with id^* and the challenge interval t^*. The challenger randomly chooses $b \in \{0,1\}$, runs **Encrypt** to encrypt M_b and receives a ciphertext C^*, which is then sent to the adversary. One restriction is that in the Extraction query the secret key for the pair (id^*, t) where $t \le t^*$ is not allowed to query.

Phase 2: The adversary will make q_i queries for $i = m+1, \cdots, n$, and the challenger responds as follows:

- Extraction query (id, j): For the pair (id, j), the same restrictions are maintained as in the challenge step, and the same response as in **Phase 1** is maintained for secret key queries.
- Decryption query (id, C, j): The triple (id, C, j) needs to satisfy $(id, C, j) \neq (id^*, C^*, t^*)$, and the challenger responds as in the first stage of query.

Guess: The adversary terminates and outputs a random bit $b' \in \{0,1\}$. If $b = b'$, we say the adversary wins this attack game.

The adversary's advantage to win the attack game is $Adv[\mathcal{A}, \mathcal{E}] = |\Pr[b = b'] - \frac{1}{2}|$ where $\mathcal{A}$ is the adversary and $\mathcal{E}$ is the scheme. Here we define the fs-sID-CCA security model. The model of fs-sID-CPA security can be obtained directly by removing the Decryption query in the query phase.

2.4. Integer lattice

Suppose $L = [\mathbf{l}_1 \mid \mathbf{l}_2 \mid \cdots \mid \mathbf{l}_m] \in \mathbb{R}^{m \times m}$ is an $m \times m$ matrix whose columns are linearly independent. The lattice $\mathcal{L}$ with rank m generated by the matrix L is defined as:

$$\mathcal{L} := L \cdot \mathbf{c} = \left\{ \sum_{i \in \{1, \cdots, m\}} c_i \mathbf{l}_i : \mathbf{c} = (c_1, \cdots, c_m) \in \mathbb{Z}^m \right\}.$$

Note that the component of $\mathbf{c}$ are integers, this paper only uses integer lattice.

For any prime q, a random matrix $A \in \mathbb{Z}_q^{n \times m}$, a vector $\mathbf{u} \in \mathbb{Z}_q^n$, set:

$$\Lambda_q(A) := \{\mathbf{e} \in \mathbb{Z}^m \text{ s.t. } \exists \mathbf{s} \in \mathbb{Z}_q^n \text{ where } A^\intercal \mathbf{s} = \mathbf{e} \pmod q\},$$

$$\Lambda_q^\perp(A) := \{\mathbf{e} \in \mathbb{Z}^m \text{ s.t. } A\mathbf{e} = \mathbf{0} \pmod q\},$$

$$\Lambda_q^{\mathbf{u}}(A) := \{\mathbf{e} \in \mathbb{Z}^m \text{ s.t. } A\mathbf{e} = \mathbf{u} \pmod q\}.$$

It is obvious that $\Lambda_q^{\mathbf{u}}(A)$ and $\Lambda_q^\perp(A)$ are q-ary lattice, $\Lambda_q^{\mathbf{u}}(A)$ can be seen as a coset of $\Lambda_q^\perp(A)$.

2.5. Discrete Gaussian and learning with errors

For a parameter $r > 0$ and a vector $\mathbf{x}$ chosen from $\mathbb{R}^n$ randomly, denote by the following function over $\mathbb{R}^n$ as

$$\rho_{r,\mathbf{c}}(\mathbf{x}) := \exp\left(\frac{-\pi \|\mathbf{x} - \mathbf{c}\|^2}{r^2}\right)$$

where $\mathbf{c}$ is the center. We note that r and $\mathbf{c}$ are omitted when the value of the two parameters equal 1 and $\mathbf{0}$, respectively. For a lattice Λ with dimension n and a vector $\mathbf{x}$ chosen from Λ randomly, the *discrete Gaussian* distribution over Λ is:

$$\mathcal{D}_{\Lambda,r,\mathbf{c}}(\mathbf{x}) := \frac{\rho_{r,\mathbf{c}}(\mathbf{x})}{\rho_{r,\mathbf{c}}(\Lambda)}.$$

Learning with Errors. Regev [24] introduced Learning with Errors (LWE), which is an intermediate problem between the worst-case problem on lattice and cryptographic constructions. That is to say, LWE problem is not only convenient for constructing public key cryptosystems, but also can be reduced to hard lattice problems in the worst case. For $q \ge 2$ and some suitable parameters n, m, and some distribution χ (discrete Gaussian distribution), the decision-LWE is stated as follows:

(1): Given the pair $(A, A^\intercal \mathbf{s} + \mathbf{e})$, where $A \in \mathbb{Z}_q^{n \times m}$ is chosen uniformly, $\mathbf{s} \in \mathbb{Z}_q^n$ is a uniformly random vector and $\mathbf{e} \in \chi^m$ is sampled from discrete Gaussian distribution.

(2): Given another pair $(A, \mathbf{b})$, where A is chosen uniformly random from $\mathbb{Z}_q^{n \times m}$ and $\mathbf{b} \in \mathbb{Z}_q^m$ is a uniformly-random vector.

The decision-LWE problem (we called $\mathrm{LWE}_{q,\chi}$) is to distinguish the two cases.

In the following theorem, Regev showed that the hardness of solving LWE in the average-case is not easier than some worst-case lattice problems by applying a quantum algorithm.

Theorem 1. *[24] Suppose $\alpha \in (0,1)$ and q is a prime which satisfies $\alpha \cdot q > 2\sqrt{n}$. If there exists an efficient (possibly quantum) algorithm that solves LWE, then there exists an efficient quantum algorithm for approximating SIVP and GapSVP on arbitrary n-dimensional lattices, in the worst case, to within $\tilde{O}(\frac{n}{\alpha})$ factors.*

Dual Regev Scheme based on LWE. In [18], the authors produced a dual Regev PKE scheme, wihch is different from Regev's cryptosystem [24]. The scheme is more suitable for the construction of lattice IBE and we briefly introduce the scheme below.

We need a shared matrix $A \leftarrow_{\$} \mathbb{Z}_q^{n\times m}$ and a function $f_A : \mathbb{Z}_q^m \to \mathbb{Z}_q^n$, where $f_A = A\mathbf{x} \,(\mathrm{mod}\, q)$ is a one-way function with trapdoor ([18], Theorem 4.9).

DualKeyGen: Sample an error vector $\mathbf{e} \hookleftarrow \mathcal{D}_{\mathbb{Z}^m,r}$ (i.e., choose the input of f_A), set $\mathbf{e}$ to be the secret key. Denote by $\mathbf{u} = f_A(\mathbf{e})$ the public key.

DualEnc: This algorithm takes $\mathbf{u}$ and a random bit b as inputs. Chooses a vector $\mathbf{s} \in \mathbb{Z}_q^n$ uniformly, outputs a ciphertext $(\mathbf{p}, C)$ where $\mathbf{p} := A^{\mathsf{T}}\mathbf{s} + \mathbf{x} \in \mathbb{Z}_q^m$, $\mathbf{x} \hookleftarrow \chi^m$ and $C = \mathbf{u}^{\mathsf{T}}\mathbf{s} + x + b \cdot \lfloor \frac{q}{2} \rfloor \in \mathbb{Z}_q^m \times \mathbb{Z}_q$, $x \hookleftarrow \chi$.

DualDec: Compute $b' = c - \mathbf{e}^{\mathrm{T}}\mathbf{p} \in \mathbb{Z}_q$. Output 1 if the value of b' is closer to $\lfloor \frac{q}{2} \rfloor$ than 0, otherwise output 0.

Theorem 2. *[18] Suppose $q \geq 5r(m+1), \alpha \leq \frac{1}{r\sqrt{m+1}\cdot\omega(\sqrt{\log n})}$ and let $m \geq 2nlog\, q$. Then the dual cryptosystem is IND-CPA secure, assuming that LWE $_{q,\chi}$ is hard.*

2.6. Basis delegation

Trapdoor Delegation. In the following theorem, the authors proposed an efficient sampling algorithm. More precisely, with some inputs n, q, the algorithm outputs a pair (A, T_A), where $A \in \mathbb{Z}_q^{n\times m}$ and T_A (a short basis of $\Lambda_q^{\perp}(A)$) is exactly in $\mathbb{Z}^{m\times m}$. The following follows from Theorem 3.2 of [3] taking $\delta := 1/3$.

Theorem 3. *[2,3] Suppose $m := \lceil 6n \cdot log\, q \rceil$. The PPT algorithm **TrapGen** takes n, q, and outputs a pair $(A \in \mathbb{Z}_q^{n\times m}, T_A \in \mathbb{Z}^{m\times m})$. Moreover, A is statistically close to a matrix which is chosen randomly from $\mathbb{Z}_q^{n\times m}$, T_A satisfies $||\widetilde{T}_A|| \leq O(\sqrt{nlog\, q})$ with negligible probability.*

Moreover, in this theorem, m is much larger than n, then the rank of outputs matrix A is n with overwhelming probability and we can hold the view that A is statistically close to a uniform matrix in $\mathbb{Z}^{n\times m}$) with rank n.

ExtBasis. To safely expand the lattice base to a higher dimensional expansion lattice, the following algorithm was introduced.

Lemma 1. *[13] There exists an algorithm **ExtBasis** taking a matrix A in $\mathbb{Z}_q^{n\times m}$ whose columns generate $\mathbb{Z}_q^n$, a basis $S \in \mathbb{Z}^{m\times m}$ of $\Lambda_q^{\perp}(A)$, and a random matrix $\overline{A} \in \mathbb{Z}_q^{n\times \overline{m}}$, the algorithm **ExtBasis** $(S, A' = A||\overline{A})$ outputs a basis $S' \in \mathbb{Z}^{(m+\overline{m})\times(m+\overline{m})}$ of $\Lambda_q^{\perp}(A')$ which satisfies $||\widetilde{S}'|| = ||\widetilde{S}||$. Moreover, the same holds even for any given permutation of the columns of A' (e.g., if columns of $\overline{A}$ are both appended and prepended to A).*

RandBasis. In [13], Cash et al. proposed a technique for randomizing lattice basis, called **RandBasis**.

Inputs: a basis T of $\Lambda_q^{\perp}(A)$, and a gaussian parameter $\sigma \geq ||\widetilde{T}|| \cdot \omega(log\, n)$.

Outputs: a basis T' of $\Lambda_q^{\perp}(A)$, which satisfies

(1): with overwhelming probability $||\widetilde{T}'|| \leq \sigma\sqrt{m}$ (m is the dimension of $\Lambda_q^{\perp}(A)$),
(2): The distributions of T' and T are independent of each other.

Preimage sampling with trapdoor. The following result was introduced by Cash et al. [12], which is a simulation tool of our construction. They showed that one can sample a preimage of f_A. Here, f_A is an one way function with trapdoor: $f_A := A\mathbf{x} \,\mathrm{mod}\, q$, where $A \in \mathbb{Z}_q^{n\times m}$ and $\mathbf{x} \in \mathbb{Z}_q^m$. The following theorem shows that given an instance $A\mathbf{x} = \mathbf{y} \,\mathrm{mod}\, q$ and a trapdoor which corresponds a submatrix of A, one can sample an preimage $\mathbf{x}$. We introduce the algorithm below.

Let $A := \left[A_1 || \cdots || A_k\right] \in \mathbb{Z}_q^{n\times km}$ where $A_i \in \mathbb{Z}_q^{n\times m}$. For $S \subseteq [k]\,([k] := \{1, 2, \cdots, k\})$, $S = \left\{i_1, \ldots, i_j\right\}$, we set $A_S := \left[A_{i_1} || \cdots || A_{i_j}\right]$, i.e., the components of A selected according to S, when $A_{i_t} \in \mathbb{Z}_q^{n\times m}\,(t = 1, 2, \cdots, j)$ is viewed as a component of A. W.L.O.G. we assume $S = [s]$ for some $s \in [k]$. The following lemma shows that under the above conditions, we can compute the preimage of f_A.

Lemma 2. *[12] The PPT algorithm **GenSamplePre**, takes a matrix $A \in \mathbb{Z}_q^{n\times km}$, a set $S \subseteq [k]$, and a random basis B_S of $\Lambda_q^{\perp}\left(A_S\right)$, a vector $\mathbf{y} \in \mathbb{Z}_q^n$, and an positive integer $r \geq \left\|\widetilde{B}_S\right\| \cdot \omega(\sqrt{\log km})$, and outputs $\mathbf{e} \leftarrow$ **GenSamplePre** $\left(A, B_S, S, \mathbf{y}, r\right)$ which is statistically close to sampling a random error vector from the distribution $D_{\Lambda_q^{\mathbf{y}}(A),r}$.*

3. Fs-ID-BTE: a binary tree fs-IBE scheme

The basic fs-IBE system that will be described in the following is fs-sID-CPA secure in the random oracle model. The reason for proposing the basic system in this section is for ease to follow. By applying FO transformation ("BCHK" transformation), we can immediately obtain a scheme which is fs-sID-CCA (IND-CCA) security in Section 4.

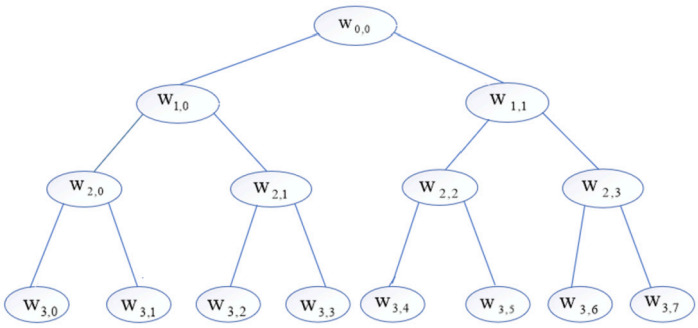

Fig. 1. Binary tree of depth $l = 4$ i.e., for $N = 8$ intervals.

The basic fs-IBE system relied on dual Regev cryptosystem which is based on the LWE assumption. Before presenting the construction of the system, we first introduce the definition of binary tree and the rules for the representation of nodes in the tree. Subsequently, we need to define some concepts that are necessary for the scheme.

3.1. Some necessary definitions and symbols

Binary tree. As showed in Fig. 1, we can define a binary tree with a depth of l. Moreover, any node of level i of the binary tree can be defined as

$$w^{(i,j_i)} = (w_{0,0}, w_{1,j_1} \cdots, w_{i,j_i})\,(0 \le i \le l-1, 0 \le j_i \le 2^i - 1).$$

The path vector shows an unique way from root $w_{0,0}$ to node w_{i,j_i}. For each $i \in \{0, \cdots, l-1\}$, we always label the nodes of this level in order from left to right, starting from 0. For any node w_{i,j_i}, we create the corresponding matrix

$$A_{w^{(i,j_i)}} = [A || A_{w_{1,j_1}} || \cdots || A_{w_{i,j_i}}] \in \mathbb{Z}_q^{n\times(i+1)m},$$

where $A \in \mathbb{Z}_q^{n\times m}$ and its associated trapdoor $T_A \in \mathbb{Z}_q^{m\times m}$ is generated by **TrapGen**. The matrix $A_{w_{i,j_i}} \in \mathbb{Z}_q^{n\times m}$ is a random matrix for all $0 \le i \le l-1, 0 \le j_i \le 2^i - 1$, and $[\cdot||\cdot]$ represents the concatenation of two matrices.

Each node w_{i,j_i} is associated with a trapdoor $T_{w^{(i,j_i)}}$, and the trapdoor could be computed by evaluating

$$T_{w^{(i,j_i)}} \longleftarrow \textbf{ExtBasis}\,(T_A, A_{w^{(i,j_i)}}).$$

In addition, the trapdoor can also be generated in another way. For example, if a trapdoor of an ancestor w_{k,j_k} of w_{i,j_i} is given, then we can compute

$$T_{w^{(i,j_i)}} \longleftarrow \textbf{ExtBasis}\,(T_{w^{(k,j_k)}}, A_{w^{(i,j_i)}}),$$

where

$$A_{w^{(i,j_i)}} = [A || A_{w_{1,j_1}} || \cdots || A_{w_{k,j_k}} || \cdots || A_{w_{i,j_i}}].$$

Minimal cover of a node. For any l-th level node $w_{l-1,t}$ $\left(0 \le t \le 2^l - 1\right)$, denoted by minimal cover MinCov $\left(w_{l-1,t}\right)$ the smallest subset of nodes that include at least an ancestor of all leaves in $\left\{w_{l-1,t}, w_{l-1,t+1}, \cdots, w_{l-1,N-1}\right\}$ and does not include any ancestor of leaves in $\left\{w_{l-1,0}, w_{l-1,1}, \cdots, w_{l-1,t-1}\right\}$. For example, in Fig. 1, MinCov $\left(w_{3,0}\right) = \{w_{0,0}\}$, MinCov $\left(w_{3,1}\right) = \left\{w_{1,1}, w_{2,1}, w_{3,1}\right\}$, MinCov $\left(w_{3,2}\right) = \left\{w_{1,1}, w_{2,1}\right\}$, MinCov $\left(w_{3,3}\right) = \left\{w_{1,1}, w_{3,3}\right\}$, MinCov $\left(w_{3,4}\right) = \left\{w_{1,1}\right\}$, MinCov $\left(w_{3,5}\right) = \left\{w_{3,5}, w_{2,3}\right\}$, MinCov $\left(w_{3,6}\right) = \left\{w_{2,3}\right\}$, MinCov $\left(w_{3,7}\right) = \left\{w_{3,7}\right\}$.

We define by $\mathcal{T}_{id,w_{l-1,t}}$ at epoch t the set which consists of the trapdoors corresponding to all nodes in MinCov $\left(w_{l-1,t}\right)$. For instance, in Fig. 1, we have $\mathcal{T}_{id,w_{3,0}} = \left\{T_A\right\}$,

$$\mathcal{T}_{id,w_{3,1}} = \left\{T_{w^{(3,1)}}, T_{w^{(2,1)}}, T_{w^{(1,1)}}\right\}$$

where $T_{w^{(3,1)}}, T_{w^{(2,1)}}$, and $T_{w^{(1,1)}}$ are associated trapdoors for $A_{w^{(3,1)}} = [A \| A_{w_{1,0}} \| A_{w_{2,0}} \| A_{w_{3,1}}]$, $A_{w^{(2,1)}} = [A \| A_{w_{1,0}} \| A_{w_{2,1}}]$ and $A_{w^{(1,1)}} = [A \| A_{w_{1,1}}]$, respectively.

Kiayias et al. [20] proposed an algorithm for efficient calculation of *best range cover* (BRC). The main idea of BRC actually includes the calculation of MinCov in this work. Taking Fig. 1 as an example, given any range $\{w_{3,i}, \cdots, w_{3,j}\}(0 \le i \le j \le 7)$, BRC can efficiently calculate the minimum nodes required to cover all leaves between $w_{3,i}$ to $w_{3,j}$. In our work, we only need to consider the MinCov of the leaves from $w_{3,i}$ to $w_{3,7}$. Therefore, we believe that the BRC mechanism can be used to calculate and implement the mechanism MinCov. For a binary tree of depth l, the minimal cover of any node contains at most l nodes, which means that the

above set of trapdoors contains at most l trapdoors. Regarding the computational complexity of the mechanism MinCov, even in the worst case it will not exceed $\mathcal{O}(l)$.

Other notions.

- In general, an id is represented by a finite string consisting of 0 and 1. For each branch of a binary tree, we always assume that the branch on the left is denoted as 0 and the branch on the right is denoted as 1. For example, we select a random node $w_{3,1}$, its corresponding string is 0001.
- We apply two hash functions: $H : \{0,1\}^* \to \mathbb{Z}_q^{n\times m}$ and $G : \{0,1\}^* \to \mathbb{Z}_q^n$. The input of H is $\left(id, w_{i,j_i}\right)$ (the concatenation of id and node w_{i,j_i}), and the function outputs a matrix in $\mathbb{Z}_q^{n\times m}$. G takes $\left(id, w_{i,j_i}\right)$ as input and outputs a vector in $\mathbb{Z}_q^n$.
- Assume the binary tree of the system has depth l. The system is specified by the depth $l(l \geq 3)$ and r (r was defined in Lemma 2). Intuitively, for an identity id and an arbitrary l, r is used to generate the id's secret key.
- We define the second leaf of the l-th level of the binary tree to correspond to the first interval in the system (i.e., we drop the first leaf). To simplify the notation, in the rest statement of this section we abuse the notation and assume that

$$w^{(l-1,t)} := t - 1 \left(1 \leq t \leq 2^{l-1}\right).$$

The first t represents the t-th node of the l-th level of the binary tree, and the latter t represents the epoch of the fs-IBE system.

3.2. Our construction

The following basic fs-IBE system is called fs-ID-BTE.

Setup: Taking as input a security parameter λ and the total epochs of the system N, do:

- Run **TrapGen** (q, n) and the algorithm and produce a pair $\left(A, T_A\right)$, where $A \in \mathbb{Z}_q^{n\times m}$ and $T_A \in \mathbb{Z}_q^{m\times m}$ $\left(T_A\right.$ is a basis of lattice $\left.\Lambda_q^{\perp}(A)\right)$.
- The system is defined over $(\mathcal{M}, C)$, where $\mathcal{M}$ is message space and $\mathcal{M} := \{0,1\}$, C is ciphertext space with $C := \mathbb{Z}_q^{ml} \times \mathbb{Z}_q$.
- Output the *params* $:= (\mathbb{Z}, n, q, m, l, r, A, H, G)$ and the $msk := T_A$.

KeyGen: Taking as input *params*, an user's id and msk, the algorithm outputs the user's initial secret key $sk_{id,0}$. To facilitate the representation of the scheme, we define some notation. For an arbitrary identity id and epoch $t \in \{0, 1, \cdots, N-1\}$ we define a matrix $A_{id,t}$ and a vector $\mathbf{y}_{id,t}$:

$$A_{id,t} = \left[A \left\| A_{id,w_{1,j_1}} \right\| \cdots \| A_{id,w_{l-1,t+1}}\right], \mathbf{y}_{id,t} = G(id, t).$$

Here, $A_{id,w_{i,j_i}} = H\left(id, w_{i,j_i}\right)$. The secret key consists of two parts: the trapdoor set and the decryption key.

- First, we build the set of trapdoors corresponding to all nodes of MinCov $\left(w_{l-1,1}\right)$ and call **RandBasis** (Since this algorithm needs to be used every time in secret key-generation, for the sake of simplicity, we assume that the parameters of the algorithm are adaptively selected and no longer show the specific details) to randomize these trapdoors. We use $\mathcal{T}_{id,0}$ to represent the set.
- Second, compute

$$\mathbf{e}_{id,0} \leftarrow \textbf{GenSamplePre}\left(A_{id,0}, \mathcal{T}_{id,0}, S, \mathbf{y}_{id,0}, r\right).$$

The secret key for the pair $(id, 0)$ is $sk_{id,0} = \left(\mathcal{T}_{id,0}, \mathbf{e}_{id,0}\right)$.

Update: Taking as input *params*, an interval t and an id, the update of $sk_{id,t}$ is as follows. It depends on the update of the matrix $A_{id,t}$, $\mathcal{T}_{id,t}$ (the corresponding trapdoors set of MinCov $\left(w_{l-1,t+1}\right)$), and the vector $\mathbf{y}_{id,t}$.

- The update of $A_{id,t}$ relies on the binary tree. The new path vector is obtained according to the path changes from root $w_{0,0}$ to each node of level-l (the nodes in level-l correspond to each epoch of N). Then cascade the components of the path vector and the id as the input to hash function H, in the way we get some new matrices, which will be recombined in the way as $A_{id,t}$ to produce the matrix $A_{id,t+1}$.
- The update of $\mathbf{y}_{id,t}$ is completed by issuing a random oracle query to hash function G.

To update $\mathcal{T}_{id,t}$, we determine the minimal cover MinCov $(w_{l-1,t+2})$. Then we derive the corresponding trapdoors for all nodes in MinCov $(w_{l-1,t+2})$. In the following we delete all keys corresponding to the nodes in MinCov $(w_{l-1,t+1}) \setminus$ MinCov $(w_{l-1,t+2})$. Also, we call **RandBasis** to randomize these trapdoors in the new trapdoor set. We compute $\mathbf{e}_{id,t+1}$ by the following algorithm,

$$\mathbf{e}_{id,t+1} \leftarrow \mathbf{GenSamplePre}(A_{id,t+1}, \mathcal{T}_{id,t+1}, S, \mathbf{y}_{id,t+1}, r)$$

where $\mathbf{e}_{id,t+1}$ is distributed over $\mathcal{D}_{\Lambda^{\mathbf{y}_{id,t+1}}(A_{id,t+1}),r}$. Outputs $sk_{id,t+1} = (\mathcal{T}_{id,t+1}, \mathbf{e}_{id,t+1})$.

Encrypt: On input *params*, a random bit $b \in \{0,1\}$, choose $\mathbf{s} \leftarrow_{\$} \mathbb{Z}_q^n$ randomly and compute $\mathbf{p} = A_{id,t}^{\mathsf{T}}\mathbf{s} + \mathbf{x} \in \mathbb{Z}_q^{ml}$, where $\mathbf{x} \hookleftarrow \chi^{ml}$. The algorithm outputs a ciphertext $\left(\mathbf{p}, c = \mathbf{y}_{id,t}^{\mathsf{T}}\mathbf{s} + x + b \cdot \lfloor \frac{q}{2} \rfloor\right) \in \mathbb{Z}_q^{ml} \times \mathbb{Z}_q$, where $x \hookleftarrow \chi$.

Decrypt: Decryption uses $sk_{id,t}$. It computes $b' = c - \mathbf{e}_{id,t}^{\mathsf{T}}\mathbf{p} \in \mathbb{Z}_q$, and outputs 1 if b' is closer to $\lfloor \frac{q}{2} \rfloor \pmod q$ than 0, otherwise outputs 0.

Correctness. By Lemma 2, it is easy to find a set S that satisfies the conditions of the algorithm **GenSamplePre**. All algorithms of the system can operate correctly since all the choices of parameters satisfy the corresponding conditions. The fs-IBE system is based on the dual Regev cryptosystem, and the correctness follows from [18], Theorem 7.1 directly.

Multi-bits extension. Given the ciphertext $(\mathbf{p}, c)$ (for message $b \in \{0,1\}$), encrypt other bit messages $\{b_1, b_2, \cdots, b_k\} \in \{0,1\}^k$ using the same method used to generate c and finally use the obtained $\{c_1, c_2, \cdots, c_k\}$ together with c to form a vector $\mathbf{c} := (c, c_{1,2}, \cdots, c_k)$ and returns the new ciphertext $(\mathbf{p}, \mathbf{c})$.

Next we give the proof of fs-sID-CPA security for the fs-ID-BTE system.

Theorem 4. *Suppose Theorem 2 holds. The dual Regev cryptosystem proposed in subsection 2.5 is IND-CPA secure. For a shared matrix A, if G and H are two hash functions, then fs-ID-BTE is fs-sID-CPA secure in the random oracle model. Further,*

$$Adv[\mathcal{A}_{BTE}, \textit{fs-ID-BTE}] \le \frac{1}{N} \cdot Adv[\mathcal{B}_{dual}, PKE] + negl(n),$$

where N is the total epochs of fs-ID-BTE.

Proof. We define $\mathcal{A}_{BTE}$ is the adversary that attacks the fs-sID-CPA security of fs-ID-BTE system. Before attacking the system, $\mathcal{A}_{BTE}$ first submits a target id^* that it will attack in **Challenge** phase. We construct a PPT algorithm $\mathcal{B}_{dual}$ that simulate the role of challenger to $\mathcal{A}_{BTE}$ and attack the dual Regev cryptosystem. $\mathcal{B}_{dual}$ first uniformly chooses $i^* \in \{0, 1, \cdots, N-1\}$. Here i^* is a guess for the challenge interval j in the **Challenge** phase.

In the following $\mathcal{B}_{dual}$ is given a matrix $A' \in \mathbb{Z}_q^{n \times ml}$ and a vector $\mathbf{u}^* \in \mathbb{Z}_q^n$ which are the main parameters of dual cryptosystem. Parse A' appropriately as

$$A' = [A || A'_{1,j_1} || A'_{2,j_2} || \cdots || A'_{l-1,i^*+1}],$$

where $A \in \mathbb{Z}_q^{n \times m}$ and $A'_{i,j_i} \in \mathbb{Z}_q^{n \times m}$ for $i = 1, 2, \cdots, l-1$ (since the path from $w_{0,0}$ to the (i^*+1)-th leaf of level-l is unique, w_{i,j_i} is a fixed node).

To simulate the challenger against adversary $\mathcal{A}_{BTE}$, $\mathcal{B}_{dual}$ sets one of the *params* to be $A \in \mathbb{Z}_q^{n \times m}$, and then sends the *params* to adversary $\mathcal{A}_{BTE}$. $\mathcal{A}_{BTE}$ may query to $G(\cdot)$ and $H(\cdot)$, we assume that

- Adversary $\mathcal{A}_{BTE}$ makes Q_H different $H(\cdot)$ -queries for some polynomial-size constant Q_H;
- Adversary $\mathcal{A}_{BTE}$ makes Q_G different $G(\cdot)$-queries for some polynomial-size constant Q_G;
- when $\mathcal{A}_{BTE}$ queries for a secret key or issues a challenge, we assume the random oracle query has completed.

Next, $\mathcal{B}_{dual}$ need to simulate the answers of any kind of queries issued by adversary $\mathcal{A}_{BTE}$. $\mathcal{B}_{dual}$ establish two empty lists, named $\mathcal{H}$ and $\mathcal{G}$, and will record the values simulated in subsequent queries.

Queries to H$(\cdot)$**.** On adversary $\mathcal{A}'_{BTE}s$ random oracle query $H(id, w_{i,j_i})$, $\mathcal{B}_{dual}$ does the following:

- If $id = id^*$ and w_{i,j_i} is one of the components of the path vector which from root $w_{0,0}$ to the node w_{l-1,i^*+1}, then $\mathcal{B}_{dual}$ returns adversary $\mathcal{A}_{BTE}$ the matrix A'_{i,j_i}.
- If $id = id^*$ and w_{i,j_i} is not contained in the path from root to node w_{l-1,i^*+1}, $\mathcal{B}_{dual}$ runs algorithm **TrapGen** to generate $A_{id,w_{i,j_i}} \in \mathbb{Z}_q^{n \times m}$ with corresponding trapdoor $T_{id,w_{i,j_i}} \in \mathbb{Z}_q^{m \times m}$. $\mathcal{B}_{dual}$ sends the matrix $A_{id,w_{i,j_i}}$ to adversary $\mathcal{A}_{BTE}$ and store the tuple

  $$((id, w_{i,j_i}), A_{id,w_{i,j_i}}, T_{id,w_{i,j_i}})$$

  in list $\mathcal{H}$.
- If $id \neq id^*$, $\mathcal{B}_{dual}$ responds in the same way as in the second case.

Note that by Theorem 3, $A_{id,w_{i,j_i}}$ is statistically close to a random matrix chosen from $\mathbb{Z}_q^{n\times m}$ uniformly.

Queries to G(·). On adversary $\mathcal{A}'_{BTE}$s random oracle query $G(id,k)$ (k is an interval), $\mathcal{B}_{dual}$ does the following:

- If $id = id^*$ and $k = i^*$, $\mathcal{B}_{dual}$ sends $\mathcal{A}_{BTE}$ the chosen vector $\mathbf{u}^*$.
- If $id \neq id^*$ or $id = id^*, k \neq i^*$, let $\mathbf{y}_{id,k} = G(id,k)$. Then returns $\mathcal{A}_{BTE}$ the vector $\mathbf{y}_{id,k}$ and store $((id,k),\mathbf{y}_{id,k})$ in list $\mathcal{G}$.

Extraction Query 1: When adversary $\mathcal{A}_{BTE}$ queries secret key of the pair (id,k), we assume that $A_{id,k}$ and $\mathbf{y}_{id,k}$ has defined.

- If $id = id^*$ and $k \leq i^*$, $\mathcal{B}_{dual}$ reports failure and terminates.
- If $id = id^*$ and $k > i^*$, $\mathcal{B}_{dual}$ first establishes the corresponding matrix $A_{id^*,k}$. In fact, we only need to ensure that $\mathcal{B}_{dual}$ can correctly simulate the trapdoor set $\mathcal{T}_{id^*,k}$. Then based on the known trapdoors, the rest part of secret key $\mathbf{e}_{id,k}$ can be calculated by **GenSamplePre** algorithm. According to the definition of binary tree, $A_{id^*,k}$ and $A_{id^*,j}$ have at least one $n\times m$ matrix different. Therefore, the trapdoor of the different $n\times m$ matrix can be simulated by **TrapGen** algorithm when answering the random oracle query of adversary $\mathcal{A}_{BTE}$. As for the other trapdoors in $\mathcal{T}_{id^*,k}$, $\mathcal{B}_{dual}$ looks up the trapdoors already recorded in the list $\mathcal{H}$ during the query phase of the hash function H and uses the **ExtBasis** algorithm to obtain them by a simple lattice basis expansion. Thus, $\mathcal{B}_{dual}$ can generate the trapdoors set $\mathcal{T}_{id^*,k}$ by using **RandBasis** algorithm and then compute the vector $\mathbf{e}_{id^*,k}$. Finally, $\mathcal{B}_{dual}$ returns $sk_{id^*,k} = (\mathcal{T}_{id^*,k}, \mathbf{e}_{id^*,k})$ to adversary $\mathcal{A}_{BTE}$.
- If $id \neq id^*$, $\mathcal{B}_{dual}$ calculates the trapdoors set $\mathcal{T}_{id,k}$ corresponding to $A_{id,k}$ by algorithms **ExtBasis** and **RandBasis** algorithms (to randomize these trapdoors). Then $\mathcal{B}_{dual}$ computes the error vector $\mathbf{e}_{id,k} \leftarrow f^{-1}_{A_{id,k}}(\mathbf{y}_{id,k})$ using the algorithm **GenSamplePre** with corresponding trapdoor $\mathcal{T}_{id,k}$ and $\mathbf{y}_{id,k}$ in list $\mathcal{G}$. Finally, $\mathcal{B}_{dual}$ returns the secret key $sk_{id,k} = (\mathcal{T}_{id,k}, \mathbf{e}_{id,k})$ to adversary $\mathcal{A}_{BTE}$.

Challenge. Assume that the adversary $\mathcal{A}_{BTE}$ decides that **Extraction Query 1** is over. Then adversary $\mathcal{A}_{BTE}$ submits a challenge (id^*, j, M_0, M_1) and $(id^*, j), (j \leq i^*)$ has never been queried in Extraction queries.

- If $j = i^*$, $\mathcal{B}_{dual}$ relays the messages M_0, M_1 to its own challenger, and obtains a ciphertext C^* from challenger, then sends C^* to adversary $\mathcal{A}$.
- If $j \neq i^*$, $\mathcal{B}_{dual}$ reports failure and terminates.

Extraction Query 2

- When adversary $\mathcal{A}_{BTE}$ queries the secret key of (id,k) and the pair are subject to the same constraints as in **Challenge**: $\mathcal{B}_{dual}$ responds as in **Extraction Query 1**.

When adversary $\mathcal{A}_{BTE}$ terminates with a random bit as the output, $\mathcal{B}_{dual}$ also terminates and outputs the same bit. From the simulation we can see that the probability of game abort is exactly $1 - \frac{1}{N}$, independently of $\mathcal{A}_{BTE}{}'s$ view. Conditioned on the simulation of $\mathcal{B}_{dual}$ is not abort, the view of $\mathcal{A}_{BTE}$ is statistically close to the real forward-secure selective ID CPA game. Indeed, the answers of the random oracle queries simulated by $\mathcal{B}_{dual}$ are indistinguishable to the answers chosen from uniform distribution (for almost all matrix A'). Particularly, the answers of the Extraction queries simulated by $\mathcal{B}_{dual}$ are indistinguishable to those output by **KeyGen** and **Update**. Finally, we claim that $\mathcal{B}_{dual}$'s advantage is exactly the same to $\mathcal{A}_{BTE}$'s if the simulation is not abort. This proves the theorem. □

Remark 1. We believe that the secret key $sk_{id,k}$ simulated by $\mathcal{B}_{dual}$ in the above proof could decrypt the ciphertext correctly. At this point, the vector $\mathbf{e}_{id,k}$ simulated by $\mathcal{B}_{dual}$ is the same as the fs-IBE system since the inputs of **GenSamplePre** are the same. Therefore, the $sk_{id,k}$ simulated by $\mathcal{B}_{dual}$ is able to decrypt the ciphertext.

4. Achieving CCA security

Boneh at al. [8] suggested that any l-level ($l \geq 1$) IND-(s)ID-CPA-secure Hierarchical IBE (HIBE) can be tightly (or efficiently) transferred into an $(l-1)$-level IND-(s)ID-CCA-secure HIBE scheme with a small extra overhead ("BCHK" transformation for short). Specifically, any IND-(s)ID-CPA-secure IBE also can be tightly transferred into an adaptive IND-CCA-secure PKE. For the construction of fs-IBE in Section 3, we immediately obtain a IND-CCA-secure fs-PKE with a small additional overhead.

Fujisaki and Okamoto [16] introduce a transformation (FO transformation for short) that can make IND-CPA-secure PKE achieves IND-CCA security, with only low additional overhead. For the construction of fs-IBE in Section 3, we immediately obtain an IND-(s)ID-CCA-secure fs-IBE. Although we have not yet achieve adaptive CCA security, we firmly believe that the selective CCA security is adequate. We have carefully considered the trade-off between security and efficiency, and achieve adaptive CCA security would necessitate further increases in parameter size and the complexity of the scheme.

5. Applications in IoT and concrete parameters

In this section, we list application examples of fs-IBE in IoT and give our parameter selection and security estimates.

Table 1
The parameter selection of our fs-IBE.

Parameter	n	q	δ	NIST Security (level)	bits security
PARA.512	512	3329	$\approx 2^{-139}$	1	143
PARA.768	768	3329	$\approx 2^{-164}$	3	207
PARA.1024	1024	3329	$\approx 2^{-174}$	5	272

Table 2
The parameter selection of our fs-IBE.

Parameter size	mpk	msk	sk	ciphertext
Our scheme	$n \cdot m$	$m \cdot m$	$\leq (m \cdot m) \cdot log\ N + m$	$m \cdot (log\ N) + 1$

5.1. Applications in IoT

Application 1: Data Transmission Security. Ensuring secure data transmission is of paramount importance in the context of the IoT. By employing fs-IBE, we can protect data during transmission, thereby guaranteeing its confidentiality and integrity. This is particularly crucial in application scenarios that involve sensitive information, such as the transmission of patient data in healthcare systems or vehicle communications in intelligent transportation systems. The forward security feature of fs-IBE ensures the timeliness of data transmission, meaning that it cannot be decrypted after its expiration. Fs-IBE prevents unauthorized third-party access, interception, or tampering of data, thereby safeguarding privacy and reliability.

Application 2: Remote Access Control. Devices within the IoT often necessitate remote access and control. Fs-IBE provides a secure mechanism for this remote access control, ensuring that only authorized users can remotely access the devices. Given the uncertainty of whether a user's system has been compromised, it is crucial to time-limit the user's authorization. This is particularly significant in home automation systems, industrial automation domains, and remote device management. If fs-IBE were implemented, it would prevent unauthorized individuals from maliciously manipulating or tampering with devices, thereby establishing a secure and reliable remote control environment.

Application 3: Secure Firmware Updates. Regular firmware updates are essential for IoT devices to address vulnerabilities, introduce new functionalities, or enhance performance. Fs-IBE ensures the security of these firmware updates, preventing unauthorized firmware manipulation or the injection of malicious firmware. This is crucial for safeguarding devices against potential attacks and misuse, especially in critical infrastructure, industrial control systems, and smart cities. Fs-IBE ensures that only authenticated and authorized firmware can be securely loaded and executed, thereby protecting devices from firmware-related security threats.

5.2. Concrete parameters

Parameter selection. Regarding the parameter selection of the encryption scheme of post-quantum cryptography, we follow the standards published by NIST. In particular, we refer to KYBER's [9] parameter selection and the NIST level 1, 3, and 5 security they achieve. Although the hard lattice problem of KYBER (MLWE) is different from the underlying hard problem (LWE) of fs-IBE in this work, KYBER's design document considers MLWE as LWE when analyzing security. Therefore, we believe that the parameter set of KYBER is our important reference.

In Table 1, we chose three sets of parameters. Given the selection of n and q, other parameters can be determined by a simple calculation, and we do not list them one by one. In the table, δ represents the probability of decryption failure, and a rough estimate of the probability of decryption failures were given. NIST security is measured by security levels, where levels 1, 3, and 5 correspond to the security of AES-128, AES-192, and AES-256 respectively. In the last column, we give specific bit security estimates.

Size of Main Parameters. In Table 2, we list the size of parameters of our fs-IBE. The parameters n, m, N in the table respectively represent the security parameter (the number of rows of matrix A in the *params*), the number of columns of matrix A ($m = \lceil 6n \cdot log\ q \rceil$) and the total epochs of fs-IBE. Moreover, previous work has neglected the consideration of other parameters in *params*, apart from the matrix A, we did not count them.

Due to the influence of binary trees, our parameters are not all "constant size" (that is, the size is independent of N). Despite this, we still achieved the same "logarithmic size" of secret keys and ciphertexts as in previous work [11,28]. Our underlying hardness assumption is LWE problem, and previous work were constructed based on the traditional hardness assumptions. Compared with the known fs-IBE scheme, the parameter size of our scheme has not increased theoretically, at least it has reached the same level.

Compared with known implementations of lattice IBE (or lattice trapdoor) [5,6,30], our parameter selection is somewhat similar, but our secret key and ciphertext size are larger than some known lattice IBE, which will affect the scheme's efficiency. In essence, this may be a trade-off between efficiency and functionality. As a compromise, using their optimization techniques may further improve the efficiency of our fs-IBE.

6. Conclusion

This paper proposed the first lattice-based fs-IBE scheme, which is selective-ID-CCA-secure and achieves forward security. Benefit from lattice, our scheme can resist quantum attacks. Moreover, we applied both the binary tree structure and the lattice structure to the construction of fs-IBE, and our parameters achieve comparatively compact size. We not only provide parameter selection and security estimates, but also propose application examples of our fs-IBE in IoT. In future work, we plan to study a new forward-secure update mechanism, which can combine lattice and fs-IBE (or fs-HIBE) without relying on binary tree. Furthermore, we will consider how to reduce the size of each parameter in lattice-based fs-IBE and design new lattice-based cryptographic algorithms.

CRediT authorship contribution statement

Renjie Jin: Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Longjiang Qu:** Investigation, Methodology, Funding acquisition, Supervision. **Rongmao Chen:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Zhichao Yang:** Investigation, Methodology. **Yi Wang:** Formal analysis, Investigation, Methodology.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

No data was used for the research described in the article.

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